

mmWave Radar Visualizer (mmWave Low Power Devices)

mmWave Radar Visualizer User Guide



Product Version: 6.0.5.1

Contents

- mmWave Radar Visualizer User Guide 1
- Contents 2
- List of Figures 3
- 1. Getting Started 4
 - 1.1. Introduction 4
 - 1.2. Quick Start Guide 5
- 2. Configuration Dashboard 6
 - 2.1. Device Connection 6
 - 2.2. Data Logging 7
 - 2.3. Configuration Selection 7
 - 2.4. Configuration Info 8
 - 2.5. CLI Command Window 9
- 3. Plots..... 10
- 4. Flash 11
- 5. Power Estimation 15
- 6. Error Handling 16
 - 6.1. ATE Calibration Unavailable 16
 - 6.2. Failed to Connect to COM port 16
 - 6.3. Error in Device Reset 17
 - 6.4. Error in Configuring the Device 17
 - 6.5. Error in Flashing 18
 - 6.6. Version Mismatch Error 18
 - 6.7. No Plots displayed / No Visualizer updates 18
- 7. Limitations 19

List of Figures

Figure 1: Home Page	4
Figure 2: Overview of Configuration Dashboard	6
Figure 3: (a) Device Connection	7
Figure 3: (b) Configuration Selection.....	7
Figure 4: Data Logging	7
Figure 5: Uploading Configuration to device	8
Figure 6: Device Configuration Input.....	9
Figure 7: Miscellaneous Plots	10
Figure 8: Overview of Flash Tab	11
Figure 9: Note on Flashing	11
Figure 10: Flashing Process	13
Figure 11: Flashing Success message	13
Figure 12: Overview of Power Estimator	15
Figure 13: ATE Calibration Error.....	16
Figure 14: COM Port Failed Error.....	16
Figure 15: Error in Halting Device	17
Figure 16: Device Configuration Error.....	17
Figure 17: Flashing Error.....	18
Figure 18: Version Mismatch Error.....	18

1. Getting Started

1.1. Introduction

The Visualizer executable (exe) file is located at <SDK_INSTALL_ROOT>/tools/visualizer/ folder with the name visualizer.exe

When the above executable is run, the Visualizer application opens up in the system default browser with the Welcome Page (Home Page) as shown below. This page provides an overview of the Visualizer and links to various documentation that can be referred to while working with the mmWave device and the Visualizer, along with short descriptions of the different application tabs.

Clicking  the button at the top right corner of the Home Page gives a quick tour on how to navigate through various sections of the Visualizer.

Clicking on the  button in the title bar will redirect to this page. The two buttons on the top right corner of the title bar,  or  provides hints to zoom the screen in or out respectively. This is useful when working with screens of different resolutions. Please note however that this shortcut may not work on all browsers, which is when the user can do a manual zoom in / zoom out from the browser settings (alternatively through appropriate keyboard shortcuts – commonly Ctrl + or Ctrl -).

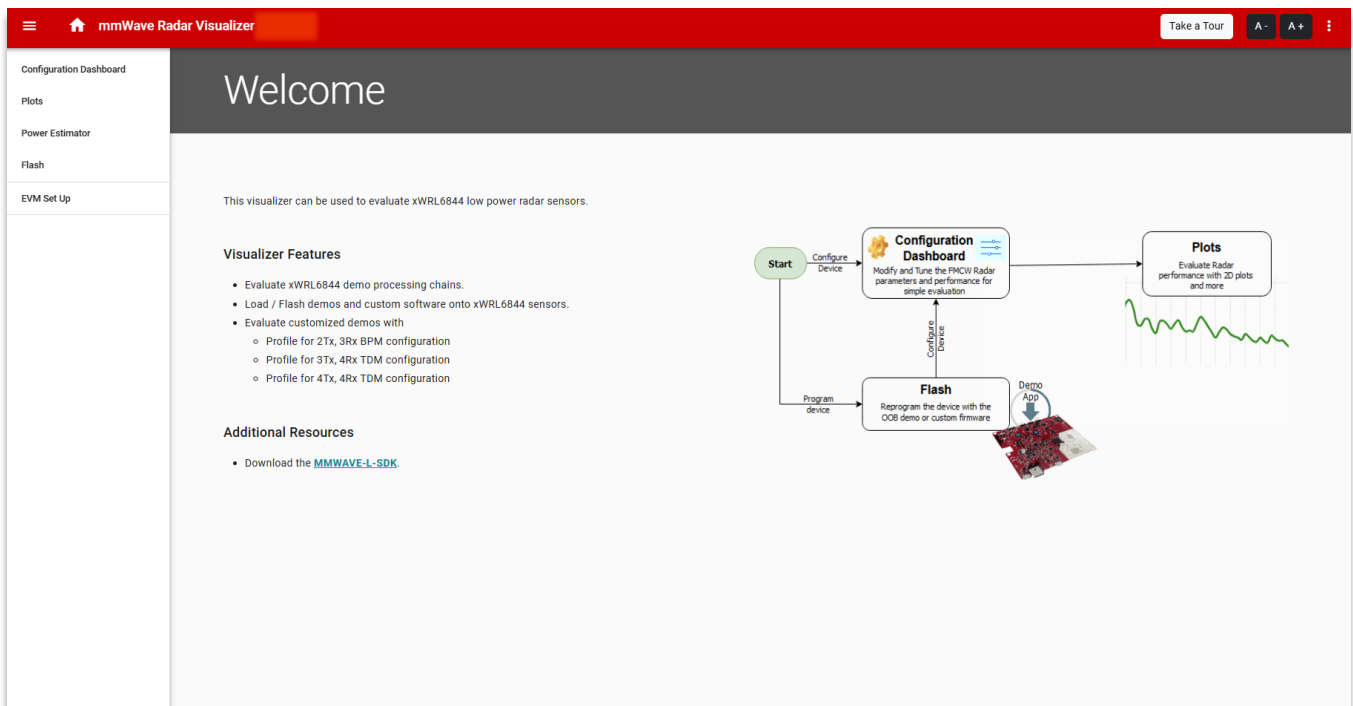


Figure 1: Home Page

The current version of the Visualizer has 3 functional tabs enunciated below

- **Configuration Dashboard** – This tab provides the functionality to modify and tune the FMCW radar parameters and performance for simple evaluation.
- **Plots** – This tab displays various plots to evaluate the radar’s performance.
- **Power Estimator** – This tab helps estimates the average power consumption for a given configuration (chirp parameters along with other inputs) and compares it against the average power reported by the device.
- **Flash** – This tab can be used to reprogram the device with the OOB demo, parking demo or any non-default firmware / custom appimage.

1.2. Quick Start Guide

1. Connect the EVM to the PC via USB.
2. Optional Step: Program a non-default firmware / app-image
 1. Navigate to the **Flash** tab.
 2. Wait for AUTO detection of COM port (else press ). Alternatively, manually select device COM port (if not already selected).
 3. Follow the subsequent steps shown in the Visualizer.
3. Configure Device
 1. Navigate to the **Configuration Dashboard** tab.
 2. Wait for AUTO detection of COM ports (else press ). Alternatively, manually select device COM ports (if not already selected).
 3. Select preset configuration under **Configuration Selection** drop-down list.
 4. Click on  .
4. **Plots** tab should display range profile and point cloud information.

2. Configuration Dashboard

The Configuration Dashboard helps configure the device and displays basic configuration information. Before configuring the device, please ensure the EVM is in `FUNCTIONAL MODE` (shown as a popup the first time when the Visualizer opens this tab). Clicking the [Take a Tour](#) button at the top right corner of the Page gives a quick tour on how to navigate through various sections of the Configuration Dashboard. The subsequent sections elucidate various elements of the **Configuration Dashboard** tab.

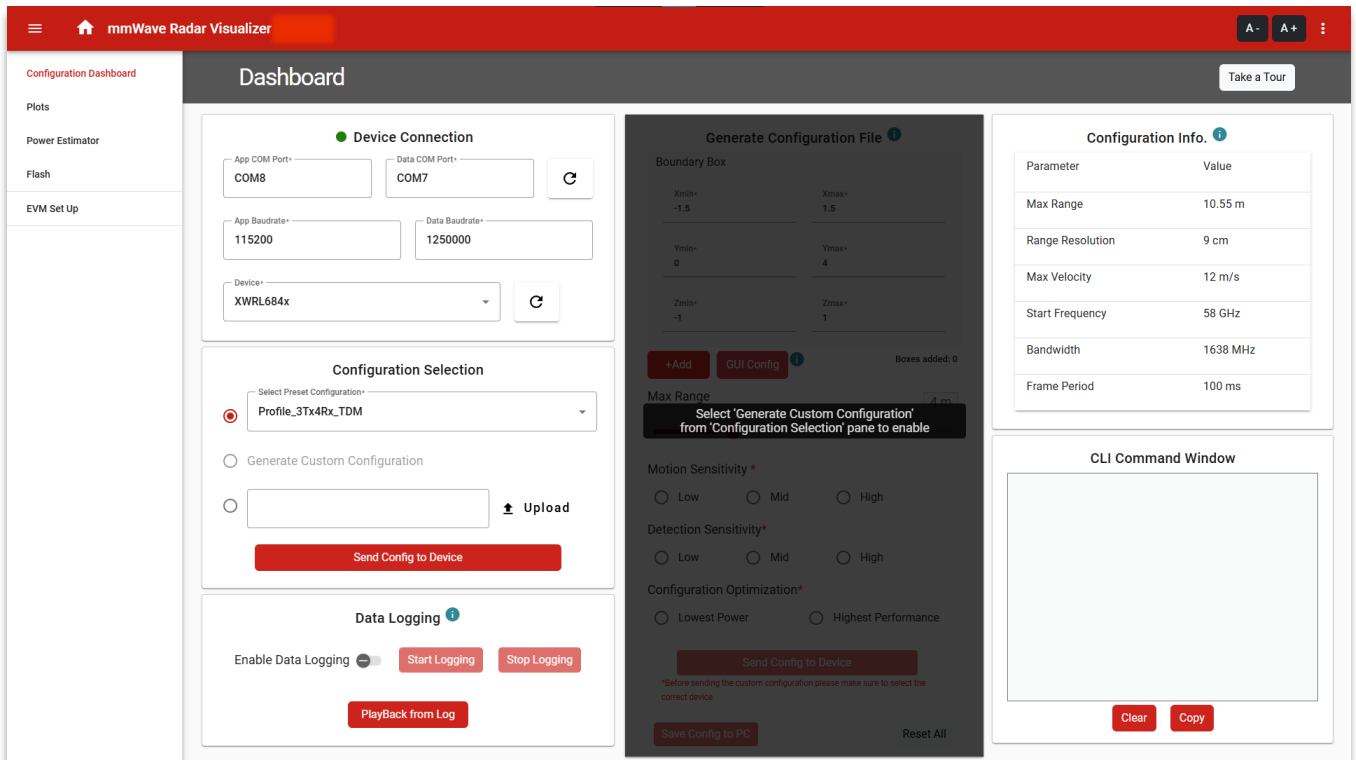


Figure 2: Overview of Configuration Dashboard

2.1. Device Connection

This section enables the selection of the Device, COM Ports and Baudrates. Please wait for a few seconds after connecting the device and opening this tab, and the COM Ports should get automatically populated. In this case, the circular symbol turns green (see [Figure 3 \(a\)](#)). If the Visualizer couldn't recognize the device COM Port the circular symbol remains red. Clicking the refresh icon fetches all available XDS110 UART COM Ports which can then be selected from the dropdown list. If the Visualizer fails to auto-detect COM Ports, please check if the system Device Manager recognizes the device connected, else enter the settings manually (may be required especially in case of a custom EVM).

Note: To know the COM ports of the device, look for the XDS110 Class Application/User UART (COMx) for the App COM Port and XDS110 Class Application Data Port (COMy) for the Data COM Port. For a device without XDS110 these could appear differently in the Device manager.

Figure 3: (a) Device Connection

(b) Configuration Selection

2.2. Data Logging

This section enables to log data output from the device. Toggling **Enable Data Logging** will capture the logs as the device starts running any configuration. The **Start Logging** and **Stop Logging** buttons can be used to dynamically start and stop capture of data output from the device respectively. In either case, a file with a name `frameData*.log` gets created under a temp folder in the same directory from where the Visualizer executable is running.

The **PlayBack from Log** button, helps select a previously generated log file (generated by the Visualizer) to replay data offline (w/o the device connected).

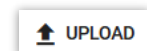
Figure 4: Data Logging

2.3. Configuration Selection

This section enables configuring the device with configuration files (*.cfg file) or with any custom configuration generated with the Visualizer. A configuration contains a bunch of CLI commands used to configure the device.

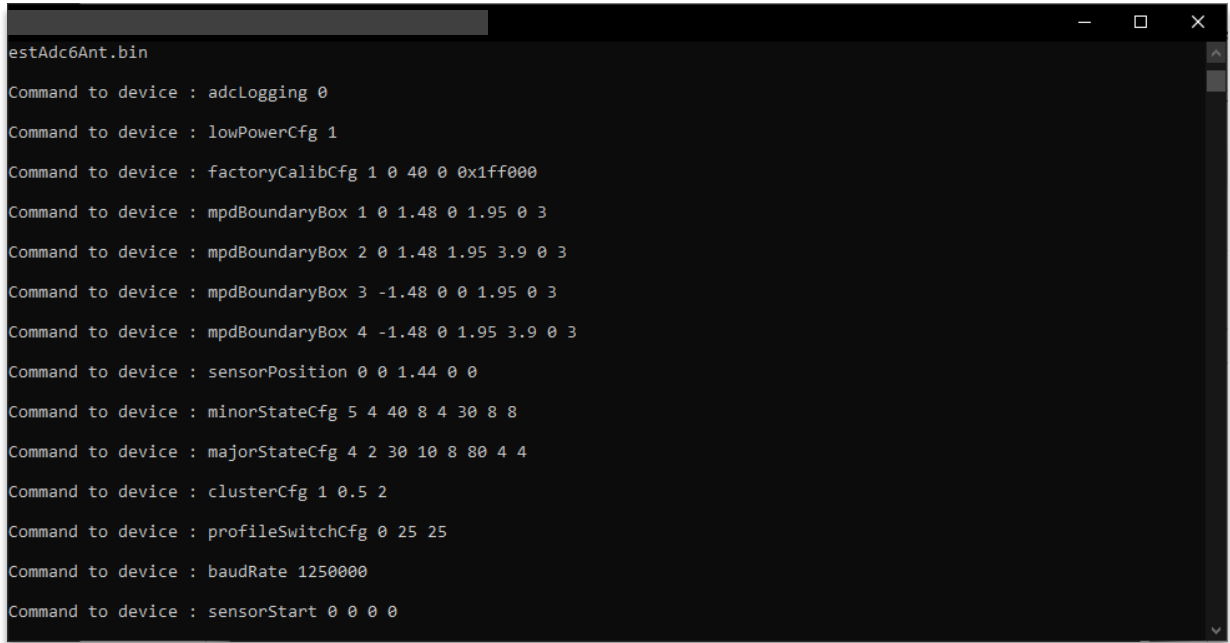
The Visualizer provides different options to configure the device (see [Figure 3 \(b\)](#)). The user may select any of the following options and then click on **Send Config to Device**.

1. **Select Preset Configuration** – Any preset configuration can be selected with this dropdown list.
2. **Generate Custom Configuration** – This option is currently not supported.
3. **Upload** – Any custom configuration can be uploaded using this option. Click **Upload** to upload any configuration file (*.cfg file) for sending to the device.



In the case of a custom device, the Visualizer will prompt with a popup to ensure that the device is reset manually.

The CLI commands to configure the device will be output on the console window as shown below as well as the **Configuration Dashboard** (see [Figure 8](#)).



```
estAdc6Ant.bin
Command to device : adcLogging 0
Command to device : lowPowerCfg 1
Command to device : factoryCalibCfg 1 0 40 0 0x1ff000
Command to device : mpdBoundaryBox 1 0 1.48 0 1.95 0 3
Command to device : mpdBoundaryBox 2 0 1.48 1.95 3.9 0 3
Command to device : mpdBoundaryBox 3 -1.48 0 0 1.95 0 3
Command to device : mpdBoundaryBox 4 -1.48 0 1.95 3.9 0 3
Command to device : sensorPosition 0 0 1.44 0 0
Command to device : minorStateCfg 5 4 40 8 4 30 8 8
Command to device : majorStateCfg 4 2 30 10 8 80 4 4
Command to device : clusterCfg 1 0.5 2
Command to device : profileSwitchCfg 0 25 25
Command to device : baudRate 1250000
Command to device : sensorStart 0 0 0 0
```

Figure 5: Uploading Configuration to device

When all the configuration commands are sent successfully, the application starts displaying the frame number (in the **Plots Tab**).

2.4. Configuration Info

This section displays config information based on the configuration selected in the **Configuration Selection** pane. The list of info. is described below.

1. **Max Range** – Maximum distance the radar sensor can detect an object.
2. **Range Resolution** – The smallest distance between two objects for them to be detected as separate objects.
3. **Max Velocity** – Maximum relative velocity of an object that can be detected by the radar sensor.
4. **Start Frequency** – Start frequency of the selected chirp configuration.
5. **Bandwidth** – Frequency bandwidth output by the device based on the chirp configuration.
6. **Frame Period** – Time in between the start of two frames.

Clicking on Sync with Device populates this section as per the current configuration of the device.

2.5. CLI Command Window

This section displays the configuration commands sent to the device.

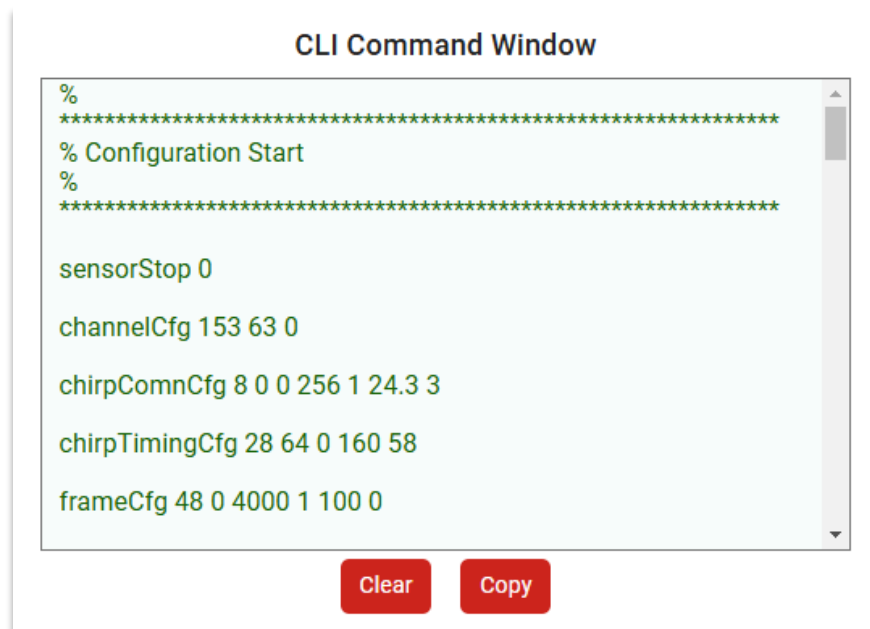


Figure 6: Device Configuration Input

3. Plots

This tab shows five important plots that can be used for further analysis viz., Range Profile, X-Y plot, Y-Z plot, Power graph and Range-Doppler Heatmap. As evident, a maximum of 4 plots can be viewed at a time. The number of Frames received and number of Points Detected are also shown.



Figure 7: Miscellaneous Plots

The `Pause` checkbox when selected halts any further updates to the visualized plots.

4. Flash

This tab enables flashing an `appimage` generated by the SDK. The device is flashed with a default `appimage`. This step is optional and allows flashing any non-default `appimage` onto the device.

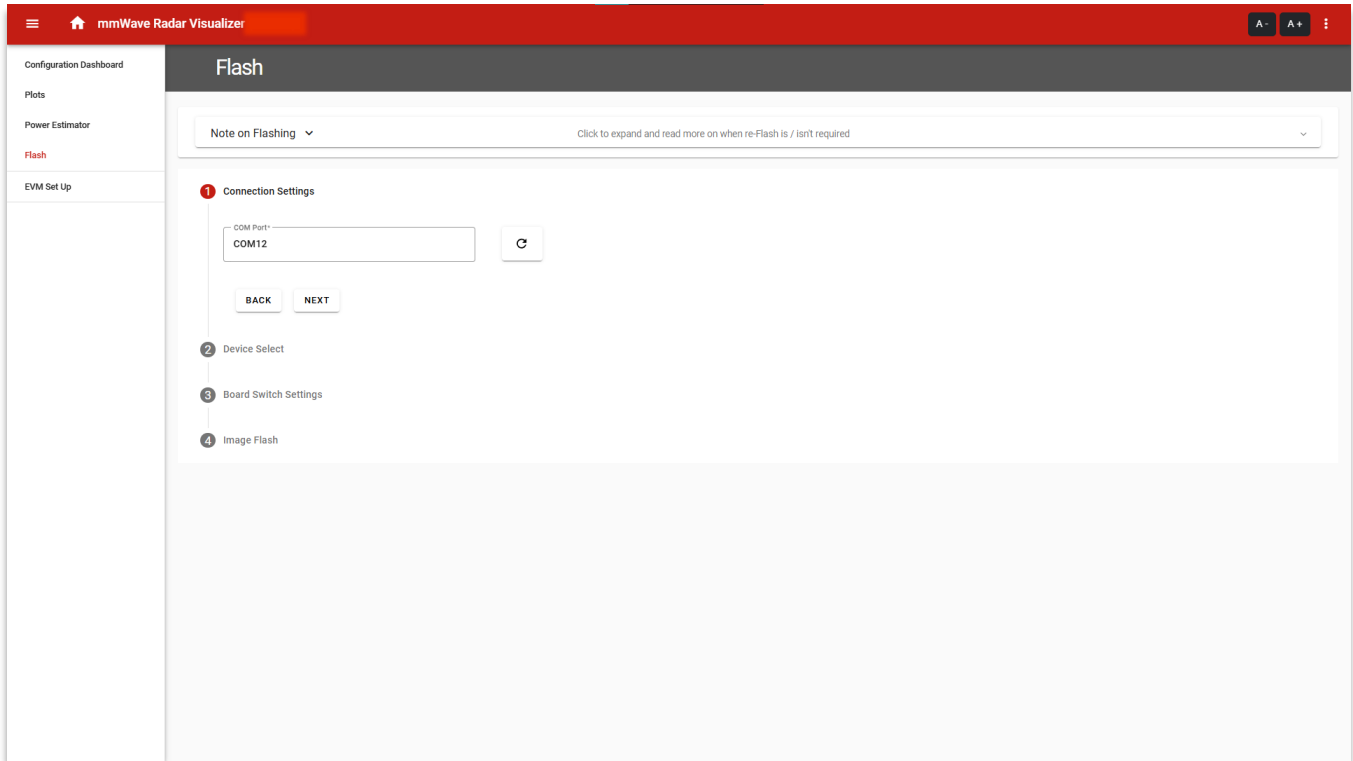


Figure 8: Overview of Flash Tab

There is a blurb at the top of Flash Tab which suggests when a reflash is/is not required.

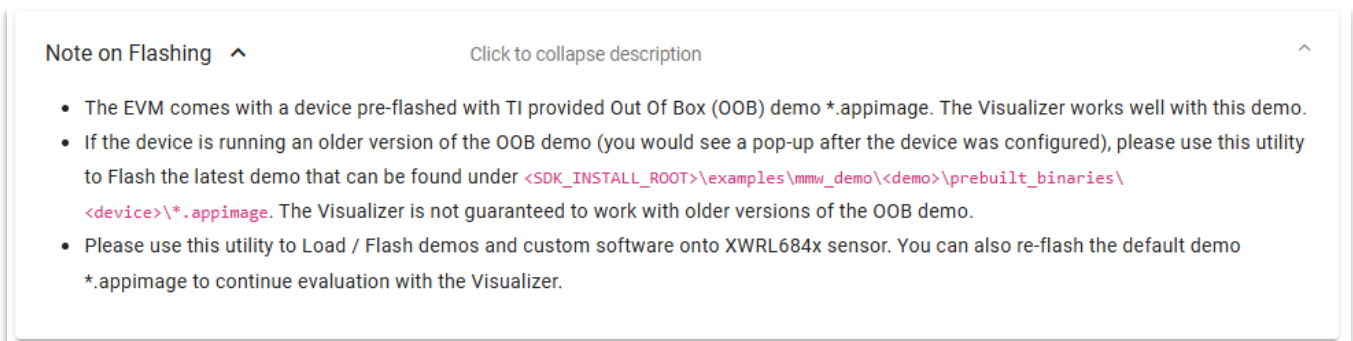


Figure 9: Note on Flashing

Click open the expansion panel to see it as shown in the above figure. Click it again to collapse it.

Flashing the device using the Visualizer involves the following steps

1. **Select COM Port** – Please wait for a few seconds after connecting the device and opening this tab, and the device COM Port should be automatically detected and populated. Otherwise, you can click on refresh icon to detect the COM Port. To know the COM Port of the device, look for XDS110 Class Application/User UART (COMx) or any other, COM Port specific to the connected EVM under the Ports (COM & LPT) of system Device manager. The COM Port can also be entered manually.

After the COM Port is selected, click .

2. **Device Select** – Select the device to be flashed with the appimage. After selecting the device click .
3. **Board Switch Settings** – The application then displays information on configuring the device. This step is mandatory, it changes the state of the device from FUNCTIONAL MODE to FLASHING MODE. After doing the necessary switch click .
4. **Image Flash** – The last step is to upload the appimage. There are three options to select the appimage to flash the device with. The first two radio buttons are enabled only when the Visualizer is run from the standard location (within the installed package) and these can be used to flash the pre-built appimage

that comes with the SDK. To upload a custom appimage, click on , select the appimage and click **OK**. Click on  to flash the appimage on to the device.

Below are snapshots of the application window and the console after clicking FLASH button.

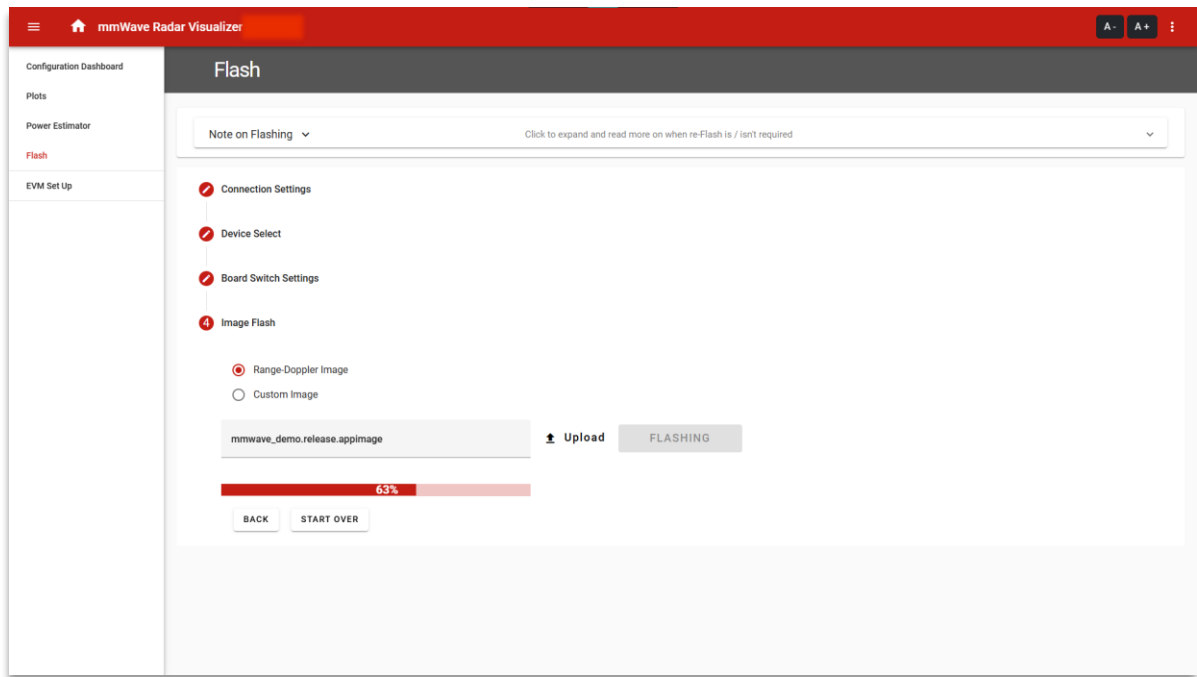


Figure 10: Flashing Process

If the Flashing was successful, a success message is shown as in the figure below.

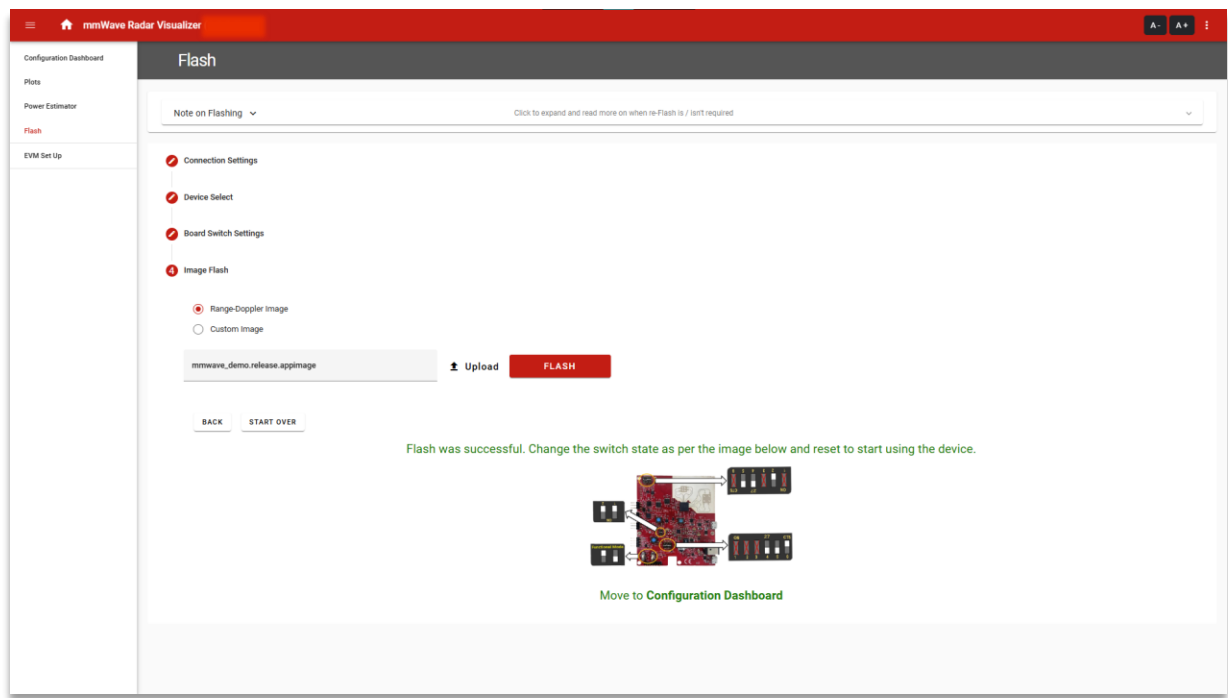


Figure 11: Flashing Success message

If the Flashing is unsuccessful, an error message will be shown instead of the Success message.

Upon reaching this step where you get the flash success message, change the switch settings of the device as per the image displayed on the application image. The device is now ready to be used in the FUNCTIONAL MODE.

Note: In case of any error shown in the console or as a popup, please ensure the device COM Port is not busy (connected to any other serial terminal or the sorts), re-connect the device, refresh the application window, check the switch settings, and repeat the above steps.

5. Power Estimation

This tab estimates the average power consumption (**Estimated Average Power**) for a given configuration (chirp parameters along with other inputs shown below) and displays it against the average power reported by the device (**Reported Average Power**).

Before switching to this tab, please ensure that the device is running a demo and is configured via the Visualizer (which is when you would see the Power Calculator Inputs pre-populated).

The number fields on the left are input fields and those on the right are output fields. The **mmWave device** should be automatically picked up by the power estimator based on the device configuration. In case it isn't, the user can select the **mmWave Device** from the drop-down list, and the **EVM Board** would then be picked up automatically.

While the Power Calculator Inputs are auto populated by the Visualizer (based on the device configuration), the user can also change the parameters to estimate the power based on custom inputs. This can be done by toggling the **Edit** switch at the top right of **Power Calculator Inputs** section. Any subsequent changes in the input fields should automatically update the output fields. The user can click Sync with Device to reset the Power Calculator inputs to match the current device configuration.

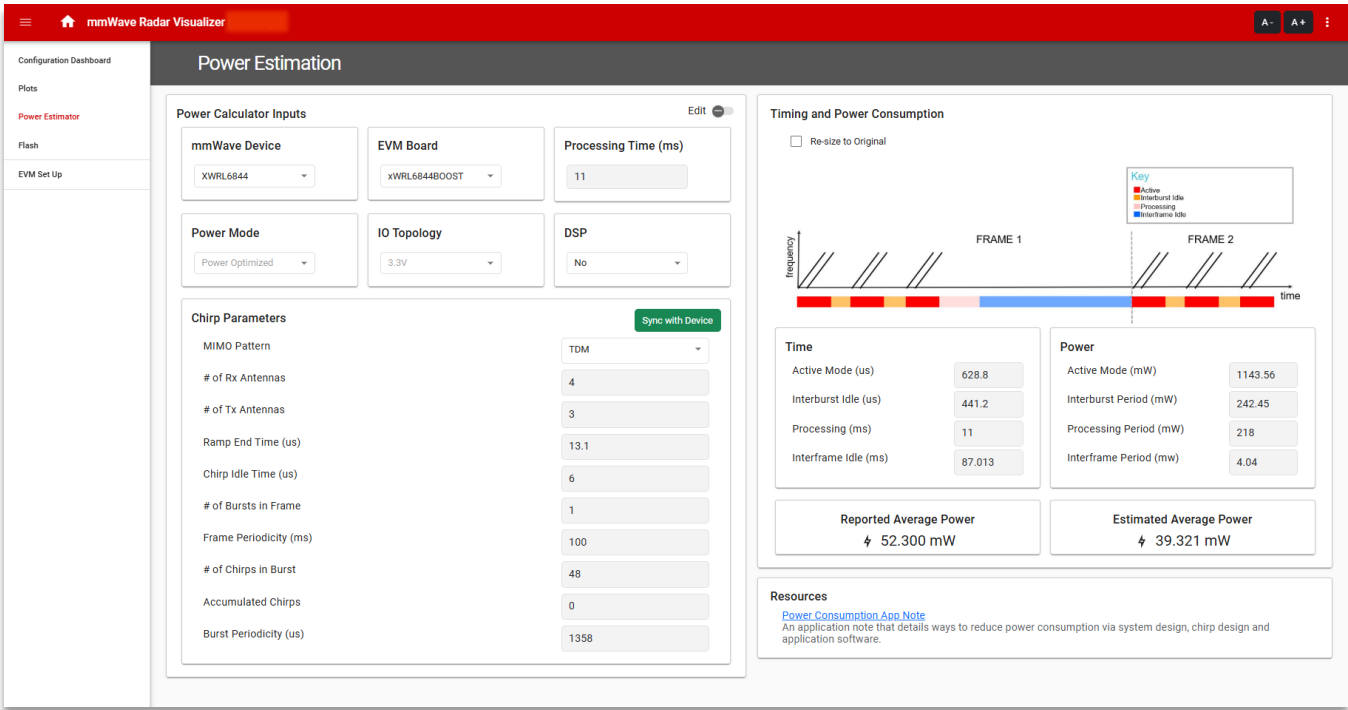


Figure 12: Overview of Power Estimator

6. Error Handling

This section details the various errors reported by the Visualizer.

6.1. ATE Calibration Unavailable

If the latest ATE calibration appimage is not flashed on to the device, this pop-up is displayed.

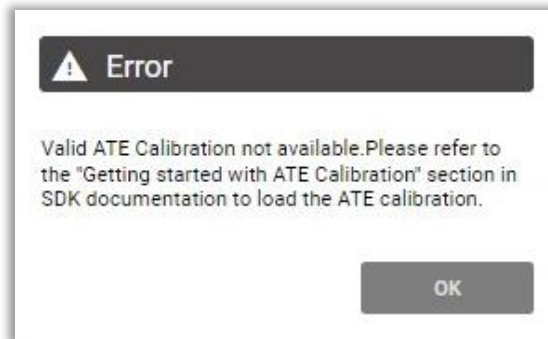


Figure 13: ATE Calibration Error

To resolve the above error, please flash the device with the latest ATE calibration appimage.

6.2. Failed to Connect to COM port

This error will mostly popup when the Visualizer is unable to connect to the device COM port. Please check if the entered COM port is correct and ensure that no other application (serial terminal) is connected to the device COM port.

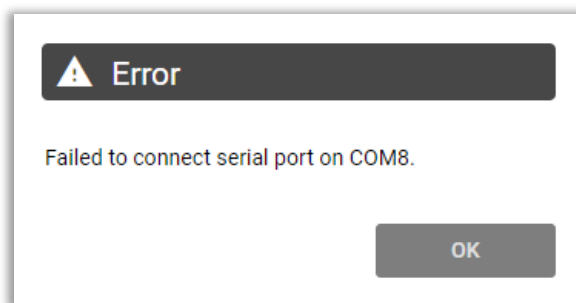


Figure 14: COM Port Failed Error

6.3. Error in Device Reset

The following error is seen when the Visualizer is unable to halt a running device. Ensure that the device is indeed connected, and if the error persists power cycle the EVM (and re-connect) and restart the Visualizer.

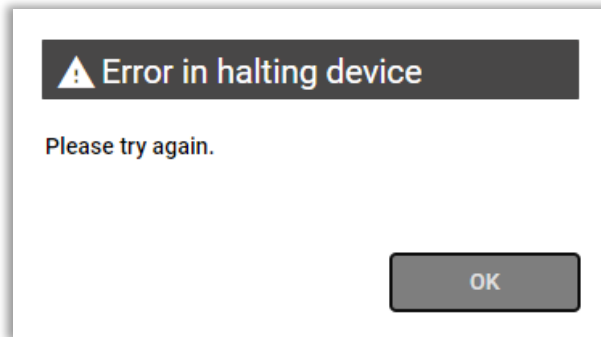


Figure 15: Error in Halting Device

6.4. Error in Configuring the Device

If the device is in `FLASHING` MODE and you try to send configuration to the device, either of the following two errors is reported.

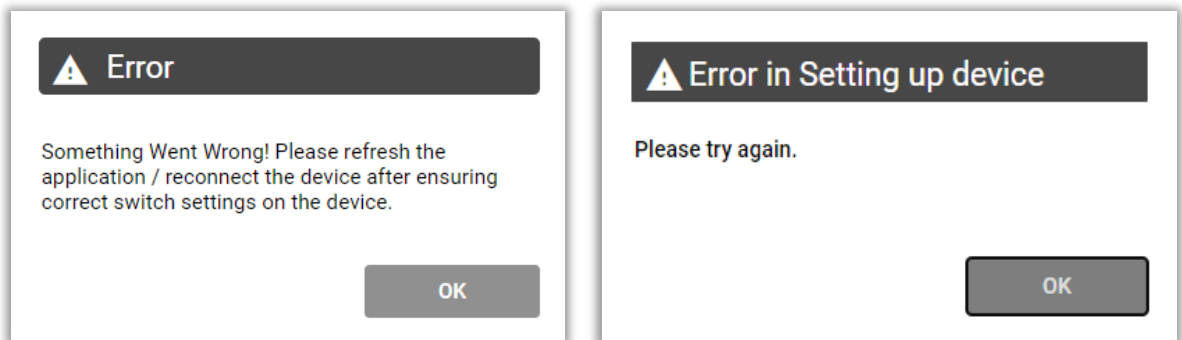


Figure 16: Device Configuration Error

6.5. Error in Flashing

If flashing the device takes too long and you see the following warning, please check if the device is switched to FLASHING MODE.

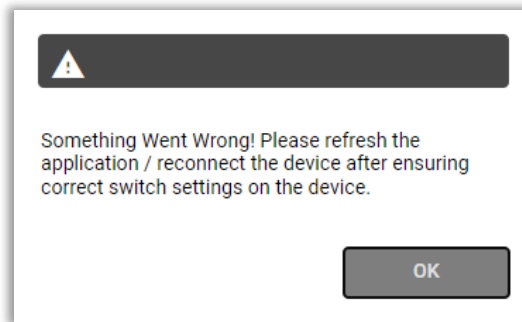


Figure 17: Flashing Error

The previous error mostly occurs when you try to flash the device that is in FUNCTIONAL MODE. It is also recommended to reconnect the USB and restart the visualizer in the event of such an error.

6.6. Version Mismatch Error

The current version of the Visualizer is supported on the latest version of the SDK it is coupled with. If the device runs an older version of the mmWave SDK demo, then the Visualizer reports an error similar as shown in the following figure.

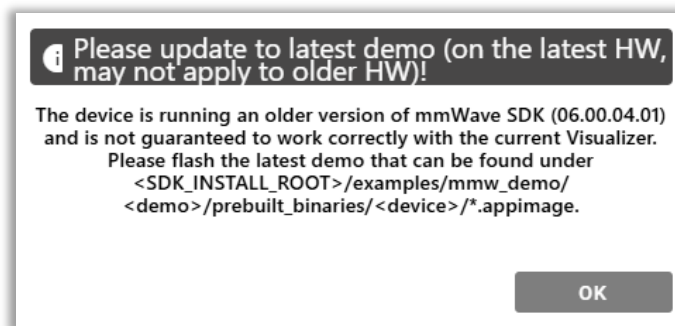


Figure 18: Version Mismatch Error

6.7. No Plots displayed / No Visualizer updates

If after configuring the device from the Configuration Dashboard, no plot is displayed, then try re-starting the Visualizer application and re-configure the device.

7. Limitations

If the device is connected after starting the Visualizer application, the `COM Port` may have to be selected / refreshed manually.

Generate Configuration File functionality is currently not supported.